

Assessing Future Availability Risk in Food Supply Chains

SupplySight - Team 16

Create X Capstone Design CS 4723-X01

•

Instructor: [Craig R Forest](#)

Mentor: [Daniel H Forsyth](#)

Georgia Institute of Technology

Atlanta, GA 30332

•

Valentina Garcia 

Vasili Fovos  (Primary Editor)

Jinho Lee 

Chenyue Wang 

Hongyi Wang 

Table of Contents

Table of Contents.....	i
Executive Summary.....	iii
Nomenclature.....	v
1.0 Introduction and Background.....	1
1.1 Problem, Motivation, and Need.....	1
1.2 Customer Discovery.....	1
1.3 Technical issues, Challenges and Opportunities.....	2
1.4 Desired solution.....	2
2.0 Existing Products.....	3
2.1 Existing Competing Products.....	3
2.2 Patent Landscape and Related Concepts.....	3
2.3 Competitive Benchmarking and Strategic Differentiation.....	4
3.0 Customer Requirements and Engineering Design Specifications.....	5
3.1 Stakeholders.....	5
3.2 Functional Requirements.....	5
3.3 Constraints.....	7
3.4 Set of Detailed Engineering Design Specifications.....	7
3.5 Relative Importance of Specifications.....	11
3.6 Overall Engineering Design Specifications.....	11
4.0 Preliminary Design Concept Ideation.....	14
4.1 Functional Requirements.....	14
4.2 Preliminary Concept Generation.....	14
4.2.1 Function 1: Data Ingestion.....	14
4.2.2 Function 2: Risk Prediction Method.....	14
4.2.3 Function 3: Risk Interpretation.....	15
4.2.3 Function 4: User interaction.....	15
4.3 Integrated Concept Development and Evaluation.....	16
4.3.1 Integrated Concept 1 – Machine Learning Dashboard System.....	16
4.3.2 Integrated Concept 2 – Machine Learning + Hybrid Dashboard/Chatbot System..	16
4.3.3 Integrated Concept 3 - Lightweight deterministic Risk Platform.....	16
4.4 Preliminary Feasibility Analysis.....	17
4.4.1 Technical Feasibility.....	17
4.4.2 Operational Feasibility.....	17
4.4.3 Resource feasibility.....	17
4.5 Preliminary Risks and Countermeasures.....	18
5.0 Concept Selection and Justification.....	20
5.1 Selection Process.....	20

5.2 Feasibility Analysis.....	21
5.2.1 Technical Feasibility.....	21
5.2.2 Operational Feasibility.....	21
5.2.3 Resource Feasibility.....	21
5.2.4 Quantitative Evaluation.....	22
5.3 Risk and Countermeasures.....	22
5.3.1 Accuracy Risk.....	22
5.3.2 User Trust Risk.....	22
5.3.3 External Risk.....	22
5.3.4 Data Risk.....	23
6.0 Engineering Analyses and Experiments.....	24
6.1 Design Analyses and Evaluation Methods.....	24
6.2 Results of Analyses and Experiments.....	25
7.0 Initial Drawing, Patent Claims, and Fabrication Package.....	26
7.1 Wireframe Representation of the User Interface.....	26
7.2 Required Sources.....	27
7.3 Patent Claim.....	27
Summary.....	29
References.....	32
Appendices.....	A-1

Executive Summary

Small to mid-size food distributors in the U.S. face persistent risk regarding future product availability, particularly for internationally sourced goods. This risk often forces purchasing decisions to be made without sufficient data, relying primarily on spreadsheets, supplier conversations, and intuition from years of experience. As global supply chains become increasingly volatile due to climate change, geopolitical uncertainty, trade restrictions and the general post-pandemic nature of economies, the financial impact of incorrect purchasing decisions has become significant. Existing enterprise systems primarily focus on demand forecasting and network-level risk, rather than short-term, product-level availability prediction.

To evaluate the validity of this problem, our team conducted customer discovery interviews with small to mid-size food distributors. From these interviews, 75% of participants identified future product availability as their primary purchasing challenge, which occurs on a weekly to bi-weekly basis in alignment with ordering cycles. This places impactful financial stress on these companies in the range of 1000 to 10000 dollars based on the severity of the shortage. This confirms the validity of the problem and the impact it causes, making a clear opportunity within this specific market segment for a viable solution

To address this need, we developed a proof-of-concept system that aggregates multiple data sources, including agricultural reports, import data, weather signals, oil prices, and industry news. These signals are processed and combined to generate a Supply Health Index (0–10) that reflects product availability for upcoming time periods. Lower values indicate higher supply risk, while higher values indicate more stable conditions.

As part of the proof of concept, we learned that combining multiple external signals provides meaningful insights into supply dynamics, which we validated with our customer. For example, changes in import volumes, abnormal price movements, and weather disruptions can be observed and aligned with supply variability. These signals are the basis for using a Random Forest Regression model to make predictions. This model is designed to capture complex, non-linear relationships between variables. Although the full model training and validation are part of ongoing work, the current system establishes the data pipeline and structure required to support this approach. The system's outputs are meant to be shown on an interactive dashboard that lets users look at the Supply Health Index and related data, allowing users to explore and study the sources of the signals. To understand the success of our platform, we will use metrics such as forecast consistency over time, easy interpretability of relevant data and inference time less than 5 seconds.

To date, the team has completed customer discovery, defined functional requirements, conducted landscape analysis, and developed a preliminary data aggregation and modeling pipeline. These results demonstrate the feasibility of the proposed approach. Next steps include expanding the data pipeline, further validating the predictive model through backtesting, and we would also like to have user testing to determine possible changes to our platform prior to final implementation. The long term goal of this product is to provide a data driven tool used by food distributors that improves decision and financial stability when related to future product availability. This will allow small to mid size food distributors operate more efficiently through uncertainty in the volatile global supply chain.

Nomenclature

API – Application Programming Interface

LLM – Large Language Model

AUC-ROC – Area Under the Receiver Operating Characteristic Curve

ETL – Extract, Transform, Load

Supply Health Index – A float with between 0 and 10, a lower value signals a high probability of a supply shortage

Cron Job – Scheduled automated task

1.0 Introduction and Background

1.1 Problem, Motivation, and Need

The Food Supply Chain is an extremely unpredictable environment. Food distributors' managers frequently have little insight into potential disruptions in supply and demand, so it's hard to make decisions about what to buy, how much to buy, and when to buy it. These decisions carry a large financial risk because food products are perishable. Margin loss, waste, spoiling, stockouts, lost sales can result from underestimating and overestimating demand.

The lack of proactive tools to detect potential availability risks in the food supply chain is the design issue this project attempts to address. Future availability risk is the likelihood that a product will become insufficient or unavailable soon as a result of external events (e.g., weather or geopolitical events), seasonality, price volatility, supplier delays, demand volatility.

Many distributors today rely on spreadsheets, ERP systems, historical sales data, and human intuition. Although these tools facilitate simple forecasting, they don't offer early warnings of new threats. With a predictive system that anticipates possible availability disruptions, distributor managers will conduct informed planning and make purchasing decisions more efficiently. To address this gap, this project proposes a system that aggregates external supply signals and translates them into a Supply Health Index (0–10) to support purchasing decisions.

1.2 Customer Discovery

We did early-stage customer discovery by finding and contacting food distributor managers, buyers, and supply chain analysts at food distributors to learn more about this problem. The purpose of these interviews was to clarify:

- How people decide what to buy right now
- What information is used to make predictions
- When there are gaps in information
- What operational or financial problems can result from missing information

Initial research indicates that:

- The margins on food distribution inventories are small, which means small mistakes make a big impact.
- There is a lot of volatility in categories like seafood and fresh fruits and vegetables.
- Forecasting systems frequently do not incorporate external risk signals.

1.3 Technical issues, Challenges and Opportunities

During the development of this system, we can encounter problems such as:

- Data integration: Combining multiple external signals such as weather, price indices, import data, and supplier-related information.
- Data quality: Historical data that is not consistent or complete.
- Forecast uncertainty: Demand and supply shocks are inherently random.
- Explainability: Procurement managers need to know why a risk flag goes up.
- Cold-start problem: There isn't any historical data on new products.

1.4 Desired solution

Our proposed solution, despite these technical problems, is to use supply chain signals to predict short-term product availability and then summarize it with a Supply Health Index (0–10). This index shows how stable supply is expected to be. Lower values mean a higher risk of disruption, while higher values mean more stable conditions. This index is made by combining data from many different sources, such as import data, pricing trends, weather signals, and news from the industry. The platform turns complicated signals into an easy-to-understand metric instead of making users figure out what different data points mean. A dashboard shows these insights by highlighting key risk factors and supporting evidence. This helps stakeholders make better decisions about what to buy.

2.0 Existing Products

2.1 Existing Competing Products

[Expana/Mintec](#) is the current leader of the agrifood market intelligence industry. They provide data and price trends on a wide range of commodities. Their platform is primarily used by large food distributors, manufacturers, and financial institutions to monitor pricing trends and inform strategies. They also survey experts in the industry to improve their LLM responses. [Vesper](#) is an upcoming competitor that uses AI to forecast commodity prices. Similar to Expana/Mintec, their solution is built on proprietary datasets and is only accessible to large enterprises. [Everstream Analytics](#) and [Resilinc](#) are supply chain risk management platforms that support network level supply chain disruption visibility. Their main customers are manufacturers but they also support monitoring for agrifood too but without any product level risk. [Gro Intelligence](#) was another agritech company that focused on using AI and public data to predict food security. Their platform supported a wide range of products but had poor PMF. They could not raise enough funds and bankrupted in 2024.

Across these platforms, the primary commercial applications include:

- Commodity price forecasting and trading support
- Supply chain risk monitoring and disruption alerts
- Procurement support for large-scale distributors
- Market analysis for financial and agricultural stakeholders

2.2 Patent Landscape and Related Concepts

As for patent search, we identified relevant concepts related to supply chain risk prediction, commodity price forecasting, and data-driven decision support systems. Existing patents focus on:

- Predictive analytics systems for supply chain disruption detection
- Machine learning models for commodity price forecasting
- Data aggregation platforms that integrate structured and unstructured sources
- Risk scoring systems that measure how much a business could be affected

These patents generally operate at a network or enterprise level, focusing on large-scale supply chain optimization or financial forecasting.

2.3 Competitive Benchmarking and Strategic Differentiation

After reviewing many different competitive products in the space, we were able to get both important details about the technical design and the strategic positioning for our product. None of the products above directly address short term product availability for food distributors but all have adjacent products that can inform our design strategy.

All the competitive products use multi source data ingestion, supervised learning techniques, and risk scoring based on a certain set of data of public data and news. This validates our idea to use structured food/agriculture from many different sources, use a model to predict the outcomes of this data, and finally present these results to our customers. Unlike all the existing products though, we are going to target a small subset of an industry, narrowing our scope down to product level availability analysis for small to mid market distributors. This will allow us to provide value to our customers that does not span the entire network level supply chain therefore allowing us to compete with products with more resources.

3.0 Customer Requirements and Engineering Design Specifications

3.1 Stakeholders

The stakeholders include:

- Primary Stakeholders
 - Buyers are small to mid size food distribution companies
 - Primary users are warehouse managers responsible for ordering inventory
- Secondary Stakeholders
 - Food manufacturers, food producers, or suppliers whose production influences supply signals
 - Customers of food distributors who depend on consistent product availability from food distributors like restaurants and grocery retailers
 - Team who works on project development
 - Advisors and Mentors that oversee project execution

Although the food distribution companies and warehouse managers are primary stakeholders, improving purchasing decisions affect the entire supply chain therefore affecting both companies upstream and downstream in this industry.

3.2 Functional Requirements

The system has several functions to assist distributors in making purchasing decisions for frozen shrimp. The functions for this are listed below:

- Data collection and preprocessing:
 - Automatically collects and refreshes external signals relevant to supply availability prediction periodically that is both monthly and daily.
 - The monthly data includes market indicators like shrimp import volumes, transportation metrics, and price signals.

- The daily data takes more frequently updated data like weather, industry specific news, oil price.
- Performance targets: Data ingestion success rate $\geq 95\%$ of scheduled runs where success rate is defined as the percentage of scheduled data pipeline runs that successfully retrieve valid data from external APIs.
- Data freshness targets: Monthly data is updated within 4 days of official release and daily environmental and news signals are updated within 24 hours.
- Baseline risk Modeling:
 - The system creates a baseline supply health index using monthly market indicators that illustrates supply conditions in the frozen shrimp market. These indicators use long-term trends in import volumes, pricing pressure, and supply volatility to predict the future availability of frozen shrimp.
 - Uses a machine learning model to predict a supply health index for the next month based on the relevant data collected.
 - Performance targets: AUC-ROC > 0.65 and a recall rate greater than 65% during backtesting.
- Daily Risk Adjustment:
 - Short-term events such as weather disruptions, disease outbreaks, policy changes, or oil price can change shrimp availability before the next monthly data update. The system uses these daily signals to adjust the baseline supply health index for users based on recent industry events. These events reflect current supply conditions within the context of the overall monthly supply trends and data.
- Reasoning:
 - To improve users ability to interpret the modeling supply health index, the system will provide relevant news sources as links and relevant market indicators in the form of charts.
 - This will allow the customer to use our system's prediction but also be able to rely on their own understanding of the data to make purchasing decisions.
- Dashboard:
 - Provide a user friendly dashboard that allows food distributors to monitor the supply conditions of shrimp and explore the signals associated with this.
 - This dashboard includes risk scoring, historical data trends, and supporting evidence in the form of news.

- Visualizations of monthly and daily metrics that allow users to easily understand current industry trends. Allows users to come to their own conclusions about the data with optional support of our model.
- Performance targets: Dashboard load time ≤ 2 seconds and supply health index update latency ≤ 2 seconds after new data ingestion.
- Security:
 - Ensure the data security and consistency from attackers and mistakes
 - Metrics: uptime $> 99\%$, backup cycle < 10 days

3.3 Constraints

The application is using only public open source data, which is daily, weekly, or monthly. This means it is hard to update the result in real-time and can only be updated with the frequency of the data. Private data, such as distributor transaction data and supplier inventory feed, cannot be included in the model, so rather than predicting the inventory of individual companies, this system focuses on providing a risk indicator. In terms of accuracy, it is impossible to predict the future with certainty since the supply chain is influenced by various factors beyond climate and historical data. Therefore the user should understand it as informational guidance instead of a certainty of prediction. The platform does not make any purchasing decisions but instead offers users more insight into the availability of products with shrimp being the first one with available data.

3.4 Set of Detailed Engineering Design Specifications

The system predicts frozen shrimp supply risk using a two-layer modeling architecture, consisting of a baseline and daily adjusted risk layer. This design reflects the way food distributors currently make decisions about how much inventory to buy based on the overall monthly market conditions and the short term daily trends.

The baseline model represents the overall monthly market conditions and captures trends in import volumes, pricing, and volatility. The specific metrics used to train our model includes the following:

Variable	Description
----------	-------------

date	Monthly observation timestamp
monthly_import	Total monthly volume of shrimp imported into the United States
avg_unit_value_per_kg	Average price per kilogram of imported shrimp
avg_air_share	Proportion of shrimp imports transported via air freight
avg_container_ratio	Ratio of containerized shipments used for shrimp transport
monthly_import_mom_pct	Month-over-month percentage change in shrimp imports
monthly_import_yoy_pct	Year-over-year percentage change in shrimp imports
monthly_import_roll3_avg	Three-month rolling average of import volume
monthly_import_roll6_avg	Six-month rolling average of import volume
monthly_import_roll3_std	Standard deviation of import volume over a three-month window
monthly_import_roll6_std	Standard deviation of import volume over a six-month window
monthly_import_zscore_6	Normalized deviation of current import levels relative to the six-month average

price_index_value	Market price index representing shrimp price trends
-------------------	---

Figure 1: Data Dictionary of Monthly data

These variables are used by the model to understand supply patterns which lead to tightening or growing supply and gives a supply health index for the next month. This data is collected automatically once per month using API requests to different databases such as the US Census Bureau for import data and the Food and Agriculture Organization for pricing data. Data is also sourced from publicly available agricultural datasets (USDA, 2023). Our system uses these databases to ingest shrimp data using scheduled scripts that run at predefined intervals once a month. Once this data is retrieved, the data is validated and stored in a structured database for use in the baseline predictive model.

Since the supply conditions change within the month due to more frequently changing data, we incorporate daily signals to adjust the baseline supply health index. These signals include environmental conditions in major shrimp-producing regions as well as global market indicators. The system collects weather variables from major shrimp-producing countries using publicly available climate datasets (NOAA, 2023). These countries represent a significant portion of global shrimp production and therefore changes in environmental conditions within these regions influence global supply and availability in the U.S.

Variables	Description
date	Daily observation timestamp
sea_surface_temp	Ocean temperature in major shrimp producing regions

precipitation	Daily rainfall levels in shrimp producing regions
wind_speed	Wind conditions affecting fishing and aquaculture operations
wave_height	Ocean conditions affecting harvesting and shipping
oil_price	Global oil price affecting shipping and logistics costs
sentiment_score	Sentiment score derived from seafood-related news articles

Figure 2: Data Dictionary of Daily data

This data is used for the daily timeframe and is acquired through the use of both external APIs and natural language models to extract indicators that are not quantitative like news. In order to make this quantitative, we rate the sentiment of every news article about shrimp as positive, negative, or neutral to extract the overall sentiment score about the product we are trying to understand.

The final supply health index is therefore derived from the combination of the monthly baseline and the adjusted daily risk. This architecture highlights the decision making process used by food distributors we interviewed as it evaluates both long-term market trends and short-term industry events when making purchasing decisions. While the supply health index is helpful, discussions with our customers indicate that our ability to aggregate the data and visualize the given data is just as important. By providing our users with the data in an easily digestible form, it allows them to come to their own conclusions about the data and does not solely make them rely on our models' conclusions.

3.5 Relative Importance of Specifications

The most important requirement for the system is the ability to provide meaningful and interpretable metrics to food distributors in order to assist them in evaluating shrimp supply conditions. This is both in the form of risk scoring and also easy to understand data visualizations.

While predictive accuracy is important, interviews with distributors indicate that explanatory information such as import and pricing trends, environmental signals, and relevant news events are also very valuable. Food distributors indicated that the algorithmic prediction would give them important guidance but would also like to see the data themselves in order to come to their own conclusions using industry knowledge.

For this reason, the system prioritizes three factors:

1. reliable automated data ingestion
2. interpretable risk predictions
3. timely updates of supply indicators with visualization of data

Daily updates allow the platform to reflect new supply signals while maintaining stability in the underlying monthly baseline model.

3.6 Overall Engineering Design Specifications

Based on the functional requirements and the engineering design specifications, the final system design must consist of automated data ingestion, predictive supply modeling, and visualizations for our data all in a unified platform for monitoring frozen shrimp supply risk.

- Data Ingestion Pipeline
 - Automatically retrieves structured and unstructured datasets from external APIs.
 - Monthly market indicators such as shrimp import volumes, price indices, and transportation metrics are collected through scheduled API queries.

- Daily environmental data for major shrimp-producing regions including India, Ecuador, Indonesia, Vietnam, and Thailand.
- News articles related to seafood supply, trade policies, and logistics disruptions are retrieved through news APIs.
- Natural language processing techniques convert unstructured news text into structured indicators such as sentiment scores and product relevance.
- Automated validation checks ensure that API responses contain required fields, correct data types, and consistent timestamps.
- Data ingestion pipelines run on scheduled intervals to continuously update the system database.
- Supply Health Index generation (anomaly detection)
 - Uses a two-layer modeling architecture with a monthly baseline model and a daily risk adjustment based on current data.
 - The baseline model analyzes monthly market indicators to capture long-term trends in shrimp supply conditions including import volumes, price, and transportation indicators.
 - The baseline model generates a supply health index representing the probability of a shrimp supply shortage within a 30 day horizon.
 - Daily environmental indicators, news sentiment signals, and overall market trends are used to adjust the baseline supply health index.
 - The adjustment layer responds to short-term disruptions such as weather events, policy changes, or logistics disruptions affecting shrimp production and trade within each month.
- Dashboard interface
 - Presents the shrimp supply health index using a user-friendly dashboard interface.
 - Displays aggregated charts and visualizations for key indicators including import volumes, price indices, environmental signals, and news sentiment.
 - Allows users to quickly identify patterns such as declining imports, rising prices, or unusual environmental conditions affecting shrimp production.
 - Provides supporting evidence with links to relevant news articles and market signals.

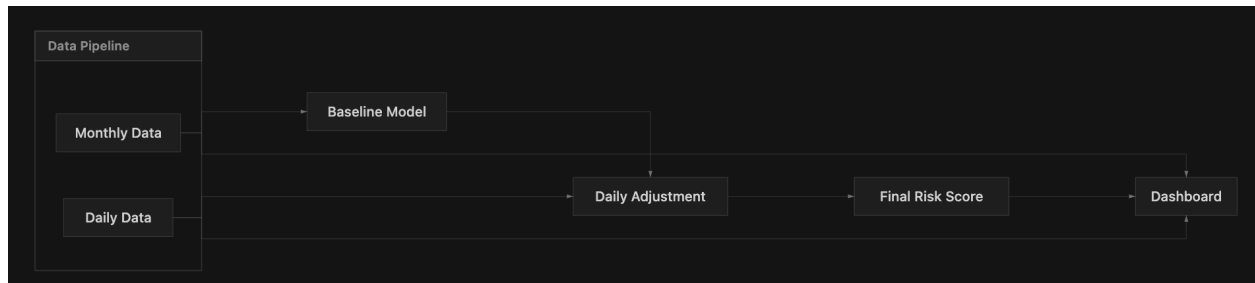


Figure 3: Data pipeline Architecture

4.0 Preliminary Design Concept Ideation

4.1 Functional Requirements

Collect open source data(USDA, weather), Normalize different data that affect the supply chain, calculate the supply health index of each product, Explain risk factor by news, Decision support assistance

The application is to provide decision-making assistance by analyzing data from the food supply chain. The model must collect and process the open source data that influences supply chain stability. Calculate each product's supply health index. In addition to providing the supply health indexes, the system explains the key contributing factors, including relevant news and events. Finally, the platform offers decision support assistance by presenting structured guidance to help users consider a purchasing strategy.

4.2 Preliminary Concept Generation

4.2.1 Function 1: Data Ingestion

- Manual Data set upload
- Automated API based ingestion from public data sources
- Hybrid approach that allows for both types for flexibility

Most likely selected: Automated API ingestion due to its scalability and reduction of manual error.

4.2.2 Function 2: Risk Prediction Method

Several approaches were considered for modeling supply chain risk:

- Multiple Linear Regression Model
- Random Forest Regression model
- Rule-based weighted scoring system

MVP Selection: Random Forest Regression model

The Random Forest Regression model is selected for the MVP because it analyzes non-linear relationships between multiple input variables, which is critical in supply chain environments where risk is influenced by factors such as import volume, price fluctuations, and climate conditions. In this system, the model takes in historical import data, price history, weather signals, and external disruption indicators (e.g., news and policy changes) to predict a continuous Supply Health Index ranging from 0 to 10. Lower scores indicate a higher probability of supply disruption, while higher scores indicate more stable supply conditions. Compared to a rule-based system, the model learns patterns directly from data rather than relying on manually defined weights, improving adaptability to changing market conditions.

Future Extension: Multiple Linear Regression can be used as a baseline model for comparison and to improve interpretability of the system outputs.

4.2.3 Function 3: Risk Interpretation

- Risk categorization (Low, Medium, High)
- Continuous numerical output (Supply Health Index)
- Supply Health Index with top driving factors

MVP selection: Supply Health Index with top driving factors to improve user interpretability. Outputs include the Supply Health Index, supporting articles, and relevant data signals that align with the prediction.

4.2.3 Function 4: User interaction

- Static dashboard
- Interactive dashboard with filtering
- Hybrid Dashboard and chatbot system

MVP selection: Interactive dashboard with filtering, as it provides a user-friendly way to explore and investigate the Supply Health Index.

Future extension: A hybrid dashboard and chatbot system where users can ask questions and better understand how the data supports the model outputs.

4.3 Integrated Concept Development and Evaluation

Using the selected functional solutions, three integrated system-level concepts were derived by combining data ingestion, modeling, interpretation, and user interaction layers into complete architectures.

4.3.1 Integrated Concept 1 – Machine Learning Dashboard System

Features of this concept:

- Automated API-based data ingestion
- Random Forest Regressor model
- Supply Health Index (0-10) + top driving factors output
- Interactive dashboard interface

In this concept, our platform will ingest public trade, climate, and other agriculture volatility data in order to generate a Supply Health Index representing expected supply stability . This index is presented through a dashboard where users can explore and investigate where the signals came from.

4.3.2 Integrated Concept 2 – Machine Learning + Hybrid Dashboard/Chatbot System

Features of this concept:

- Automated API-based data ingestion
- Random Forest Regression Model
- Supply Health Index (0-10) + top driving factors output
- Interactive dashboard interface

In this concept, our platform will do all the same functional things as concept 1, but we will add a chat system where users can ask questions to gain more insight about the Supply Health Index and the factors driving it. This will give users more confidence and trust in the data and reasoning our model is using.

4.3.3 Integrated Concept 3 - Lightweight deterministic Risk Platform

Features of this concept:

- Automated API-based ingestion
- Rule-based weighted scoring

- Supply Health Index output
- Static dashboard

This concept eliminates machine learning and relies on predefined feature weights to compute composite Supply Health Index. This would be the most simple and highly feasible but, it lacks a learning model that can adapt to nonlinear interactions among supply indicators.

4.4 Preliminary Feasibility Analysis

4.4.1 Technical Feasibility

Concepts 1 and 2 are technically feasible given that:

- Available structured public data (USDA, NOAA, import data, and news sources)
- Established machine learning libraries for our use cases supported by research (Xu et al., 2024)
Defined prediction targets and measurable input signals for training and evaluation of the Supply Health Index
- 8-week development timeline with 5 team members who are all computer science majors

4.4.2 Operational Feasibility

The proposed system does not require any integration with the existing systems that food distributors already use for inventory management. This means that our platform can stand independent from their other software and act as a separate decision support layer which companies use in collaboration with existing systems.

4.4.3 Resource feasibility

Development workload distribution:

- Data pipeline engineering
- Feature engineering and labeling
- Model development and validation
- Dashboard front-end development
- Integration and testing

The scope is manageable within the semester for five developers

4.5 Preliminary Risks and Countermeasures

- User trust risk:

If the evidence is not explained, users may not feel reliable about the supply health index. Simple numerical output can limit operational use. In the B2B decision-making process, the user's trust affects system reliance. It needs to provide interpretable insights to support managerial decision-making (Baryannis et al., 2019)

Countermeasures: The system describes the main factors that influence supply health index. It also displays related news articles and data along with visualizations. This enhances transparency.

- Data update risk:

Public data sources such as USDA reports, NOAA climate data, and trade statistics are updated at different frequencies. Different update cycles can reduce responsiveness.

Countermeasures: System will use time alignment logic to calibrate the update cycle. Also it will display the timestamp of the last update for transparency.

- Project schedule risk:

8 weeks might not be enough time to implement all the features. (machine learning pipelines, data engineering processes, dashboard interfaces, and chatbot functionality) It increase the risk or incomplete integration.

Countermeasures: Using phased prioritization strategy. A minimal functional pipeline(data engineering processes, dashboard interfaces) will be implemented first for a working prototype. Advanced features(chatbot) will be a stretch goal. It will decrease the schedule risk.

- External dependency risk:

As most of the data is from external sources, API changes, API failure, API error, or access restriction can occur. It makes the system unstable, which leads to degraded user experience and lower reliability.

Countermeasures: The system will store all processed data into an internal database to establish a fallback dataset. If an API failure occurs, the system will calculate a supply health index based on

collected data. Additionally, a local caching mechanism will be implemented to deal with temporary API errors. Modular Architecture will be used so that even if the API changes, this will allow easy modification.

5.0 Concept Selection and Justification

5.1 Selection Process

Evaluation of each concept was necessary to determine the appropriate concept. As mentioned earlier, concept 1 is the Machine Learning Dashboard System, concept 2 is the Machine Learning + Chatbot System, and concept 3 is the Rule-Based Risk Platform. Three different concepts provide different solutions to the distributor's supply uncertainty problem in various directions.

The concept selection process applies the requirements obtained through customer discovery and stakeholder analysis. The system should make it easy to see the data that the stakeholders need at a glance, such as news, export, and oil price, and should be automatically updated at least daily. It should not only collect and show the data, but also interpret and suggest the options.

A decision matrix approach was used to efficiently compare design concepts. The criteria are as follows. Each criterion was weighted according to importance and rated 1 (bad) to 5 (good).

Concept selection will be discussed based on three criteria: Predictive reliability - Reliability is essential because the purpose is to help in the decision-making process. Visualization or data collection alone cannot simplify the user decision process, explainability - It needs to explain the data and express the underlying raw data. Food distributors, the main target group, have gone through an intuitive decision-making process so far, so they need a sufficient explanation, implementation feasibility - Prototypes should be made by 5 CS undergrads in 8 weeks.

Criteria	Concept 1: ML Dashboard	Concept 2: ML + Chatbot	Concept 3: Rule-Based Platform
Predictive reliability	5	5	2
Explainability	5	4	2
Implementation feasibility	4	2	5
Total Score	14	11	9

Figure 4: Decision matrix

Concept 1 scored the highest, so it was selected as the final concept. It is possible to visualize data and present opinions, which are core features, while maintaining high reliability and feasibility. It is the most balanced option.

5.2 Feasibility Analysis

5.2.1 Technical Feasibility

The selected concept uses a reliable public data source, a verified machine learning model. NOAA and USDA provide its data through API. The data ingestion pipeline has already been implemented and tested.

Random forest models are the most suitable for analyzing supply chains because it can capture non-linear relationships between multiple variables (Xu et al., 2024).

The interface will be implemented through a web library. According to performance benchmarks, data processing and model execution meet the latency requirements of the system. The system processes 365 days of data in approximately 4.7 seconds, and anomaly detection executes in under 100 milliseconds, indicating efficient system performance.

5.2.2 Operational Feasibility

The concept can be used independently without the need for separate integration with the systems currently in use by food distributors. This improves accessibility and eliminates the need to modify current workflows. It just simplifies procedures and saves time.

The system is consistent with the decision-making process that relies on real-time data currently used; it simplifies current decision-making processes by visualizing the data and showing conclusions.

The interface is designed to be intuitive, allowing even non-technical people, such as warehouse managers, to understand the data, which reduces training costs.

5.2.3 Resource Feasibility

Feature engineering, model implementation, and dashboard design remain. Data pipeline development is in the final stages. Existing libraries allow teams to focus on optimization.

In addition, we will focus on key features development, such as data ingestion and baseline modeling. Other features will be considered for future expansion.

5.2.4 Quantitative Evaluation

Quantitative analyses were conducted. In our internal testing the system processes one year of data in less than a minute and can handle the amount of data we are currently aiming for locally without an external server. This performance demonstrates that the system can efficiently scale within the intended deployment scope. Concept 1 provides a better balance between performance, interpretability, and implementation feasibility.

5.3 Risk and Countermeasures

5.3.1 Accuracy Risk

The accuracy of the learning model. Even if the data and the data processing process are perfect, it cannot predict the future. It is difficult to guarantee the accuracy of future predictions. In addition, it takes time to test the accuracy.

Therefore, we will backtest using historical data. Also, use a model that fits the supply chain's analysis. These ensure the minimum accuracy of the system.

5.3.2 User Trust Risk

Providing accurate data and predictions is useless if users don't believe them. Users distrust the results if they don't know the source or are unfamiliar with it. This can happen in a retail industry with a traditional atmosphere.

Visualizing raw data and news, along with predictions, makes users understand the process and background through which conclusions are made. This makes the users trust. Also, an easy design makes it less accessible.

5.3.3 External Risk

Supply chains are affected by a variety of factors. Not just supply issues, but also geopolitical issues, resulting in several unusual events. This causes uncertainty in the system.

The system is positioned as a decision support tool. The platform is designed to provide insights and evidence to help users in the decision-making process. This frees the platform from errors caused by uncertainty.

5.3.4 Data Risk

Data depends on public data, which leads to situations in which the system becomes inoperable when problems occur, such as API updates.

The system stores the data in the internal database to produce output using data from the internal database when an issue occurs.

6.0 Engineering Analyses and Experiments

6.1 Design Analyses and Evaluation Methods

To ensure the SupplySight architecture meets our specifications for accuracy, usability, and reliability, we are conducting two types of design analysis:

- Analytics Engine Latency Analysis
 - We analyze the execution time required to ingest data from the Open-Meteo Marine API, calculate historical baselines, and execute the Z-score calculation. This ensures our backend can scale as we include additional metrics and features.
- Model Performance Analysis
 - We are evaluating the predictive accuracy of our Supply Health Index by comparing results from a Random Forest Regressor against a baseline Multiple Linear Regression model. We use evaluation metrics such as Mean Squared Error and R-squared values based on historical import and pricing data. This ensures that we are ensembling the models with appropriate weights to produce a supply health index that accurately reflects the reality.

We evaluate our software design iteratively with our target customers: food distributors and supply chain managers.

- Latency Requirements
 - The core function of the software is to provide daily strategic insights. Since the weather, news, and geopolitical data updates on a daily basis, millisecond latency is not necessary for the backend data ingestion. Users of our product interact with the frontend dashboard often. The calculated insights must be loaded with sub-500ms latency and cached; relevant news with summary and url must be loaded under 2.0 seconds to ensure a seamless user experience.
- Defining Success
 - The success of the software is quantified mathematically by the reduction of mean squared error in our supply predictions, and user-centric metrics, such as reported cost saved and reduction in a distributor's time spent on supply chain research.

6.2 Results of Analyses and Experiments

- Analytics Engine Benchmarks
 - We tested the data pipeline to evaluate its ability to pull historical data, calculating daily averages from hourly data, filling in missing data from previous values, and writing to a Postgres database. We tested the anomaly detection module to evaluate its ability to pull data of specific ranges from a Postgres database and calculate Z-scores of different fields simultaneously.
 - The data pipeline successfully ingested and processed 365 days of weather, wave, and geopolitical data across 5 target regions in 4.7 seconds. The anomaly detection module accurately calculated and flagged statistical anomalies from a sample of 365 days in 88 ms. The performance shows that the functions are operating within the acceptable timeframes.
- Models Performance Benchmark
 - We will be training a Multiple Linear Regression model to establish a baseline and extract feature importance, followed by a Random Forest Regressor to capture non-linear relationships in the weather and pricing data.

7.0 Initial Drawing, Patent Claims, and Fabrication Package

7.1 Wireframe Representation of the User Interface

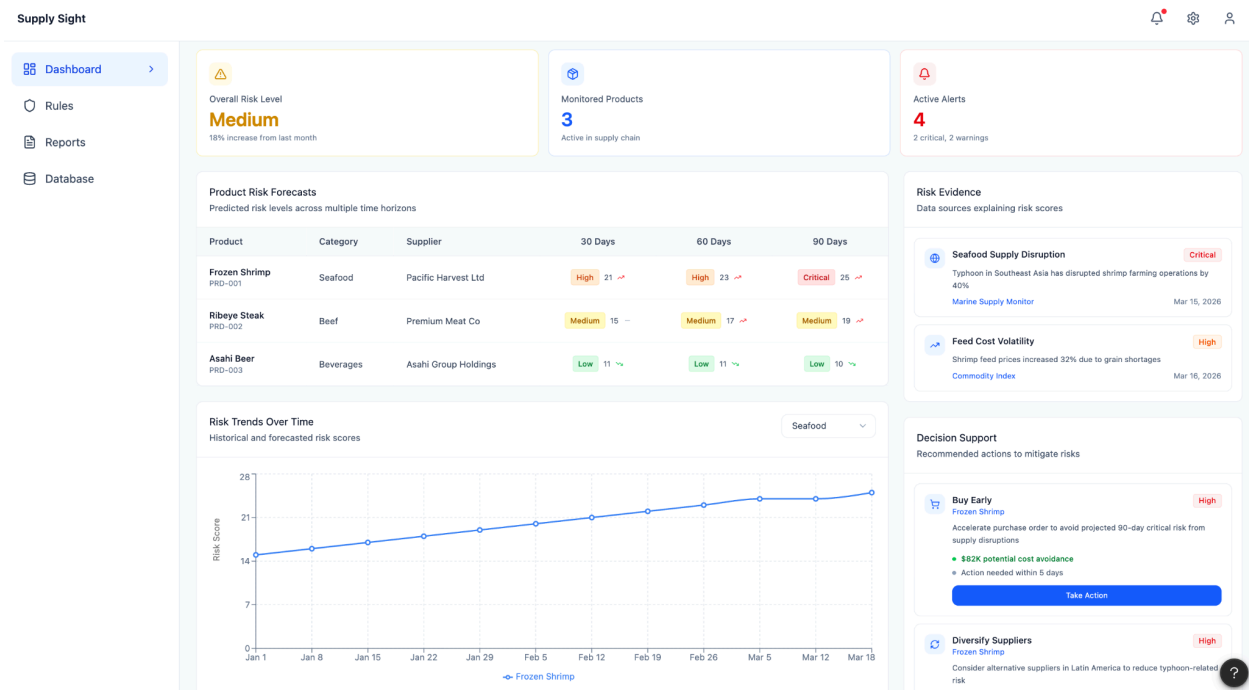


Figure 5: Main Dashboard Wireframe

This interface is a centralized dashboard that displays product-level supply risk insights. The layout includes:

- A navigation bar and sidebar for accessing different modules (dashboard, rules, reports, database)
- A risk overview section showing key metrics such as overall risk level, number of monitored products, and alert counts
- A product risk table displaying predicted risk levels across multiple time horizons (30/60/90 days)
- A trend visualization panel that illustrates changes in risk over time
- An evidence panel that provides explanations for risk scores using relevant data sources and news

- A decision support panel that recommends actionable strategies (e.g., buy early, delay purchase, switch suppliers)

7.2 Required Sources

As a software-based system, the bill of materials consists of the required data sources, infrastructure, and software tools needed to build and run the system.

The system uses several types of data sources. These include but not limited USDA agricultural data such as production and import/export statistics, NOAA weather data including temperature and precipitation, public trade datasets that show import and export flows, and news data collected through public APIs.

For infrastructure, the system relies on a cloud provider. It uses a database to store data. Computation is handled through cloud virtual machines, and data storage is managed using cloud storage services.

On the backend, Python is used for data processing and modeling. For machine learning, libraries such as Scikit-learn, numpy, Prophet, and etc are used to implement models like Random Forest. On the frontend, javascript is used to build the dashboard interface, along with charting libraries such as Chart.js.

The estimated total cost for a minimum viable product is relatively low. Cloud computing costs are expected to be below \$30 per month, while API usage for language models and data services may cost about \$50 per month. Overall, the total estimated monthly cost is about \$80.

7.3 Patent Claim

This invention is a computer based method for predicting supply availability risk at the product level within a supply chain. The system collects and combines multiple types of external data and uses a machine learning model to generate a risk score that predicts the likelihood of a future shortage over a given time period. It also identifies and ranks the key factors that contribute to this risk. The results are presented through a user interface that shows both the risk score and supporting explanations, along with recommended actions. Users can interact with the system by adjusting parameters, running simulations, and overriding recommendations.

The novelty of this invention mainly focuses on combining multiple types of external data, both structured and unstructured, to produce product-level supply risk predictions. Unlike traditional systems, it also provides explanations tied to real-world events such as news and data changes, and includes interactive tools that allow users to explore different decisions.

Summary

The team is continuing to work to solve the supply chain uncertainty in the food distribution industry. Through additional interviews and data they provided (e-mail between distributors and suppliers), we found out that import and export data, environmental information, price indicators, news, and oil prices are factors that influence the current decision-making process. Therefore, we collected the data and confirmed that the data is in the form of a public api and Excel data sheets exist.

For the prototype, the system architecture was designed. In order to connect and train data to the machine learning model, a data pipeline that processes various data sources was implemented and tested. Data processability and system efficiency were confirmed through performance tests.

We've got enough information from our customers, so our next step will mainly focus on prototypes. Training the model and backtesting using historical data is scheduled. Our main goal is to connect models, improve accuracy, and stability.

Internally, the team updates the current status through regular meetings every week and divides tasks among individuals. Since continuous communication is maintained even after the meeting through group chat, the response is fast when a problem occurs.

The development status will be continuously shared with customers. Since the basic structure has been confirmed, the focus will shift toward discussions on the dashboard interface. and after the development is completed, we will also proceed with prototype testing together.

We have a weekly meeting with the mentor to discuss the progress and the problems we are facing.

Gantt Chart

Task	Start Date	End Date	Duration
Define risk scope and success criteria	Feb 17	Feb 19	3
Choose datasets (USDA, weather, news)	Feb 18	Feb 21	4
Design pipeline architecture diagram	Feb 20	Feb 24	5
Design database schema	Feb 21	Feb 25	5
Define LLM JSON output format	February 24	Feb 26	3
Build USDA ingestion pipeline	March 1	Mar 8	8
Build weather ingestion pipeline	March 3	Mar 10	8
Set up news data collection	March 5	Mar 11	7
Implement LLM extraction for news events	March 8	Mar 15	8
Store structured + extracted data in DB	March 10	Mar 16	7
Define features for risk model	March 18	Mar 21	4
Prepare training dataset from DB	March 20	Mar 24	5
Train first ML model prototype	March 25	Mar 30	6
Evaluate model and refine features	March 25	Apr 3	6
Implement risk score output logic	April 2	Apr 5	4
Connect model to database pipeline	April 4	Apr 7	4
Build API or script to serve predictions	April 6	Apr 9	4
Test end-to-end pipeline	April 8	Apr 11	4
Design dashboard layout	Apr 10	Apr 12	3
Build dashboard frontend	Apr 12	Apr 18	7
Connect dashboard to backend/API	Apr 15	Apr 20	5
Add alerts and explanation features	Apr 18	Apr 22	4
System testing and debugging	Apr 18	Apr 23	6
Prepare presentation slides	Apr 20	Apr 23	4
Record demo backup/screenshots	Apr 22	Apr 24	2
Final rehearsal and polish	Apr 23	Apr 25	2

Figure 6: Gantt Chart in table

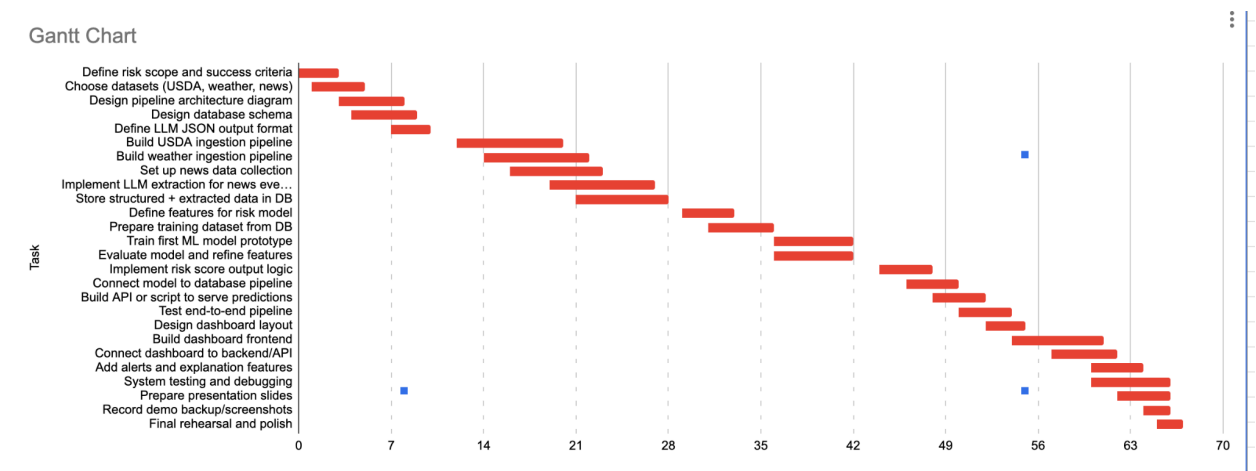


Figure 7: Gantt Chart

References

Baryannis, G., Validi, S., Dani, S., & Antoniou, G. (2019). Supply chain risk management and artificial intelligence: state of the art and future research directions. *International Journal of Production Research*, 57(7), 2179–2202. <https://doi.org/10.1080/00207543.2018.1530476>

National Oceanic and Atmospheric Administration. (2023). *Climate data and weather anomalies*. U.S. Department of Commerce. <https://www.noaa.gov>

United States Department of Agriculture. (2023). *Economic Research Service: Data and reports*. U.S. Department of Agriculture. <https://www.ers.usda.gov>

Xu, Z., Ni, W., Liu, S., Wang, F., & Li, J. (2024). Logistics supply chain network risk prediction model based on intelligent random forest model. *IEEE Transactions on Engineering Management*, 71, 9813–9825. <https://doi.org/10.1109/TEM.2023.3317922>

Appendices

Appendix A – Academic Support for Random Forest Model Selection

This appendix summarizes the academic foundation supporting the selection of a Random Forest based supervised learning model for supply chain risk prediction.

Xu et al. (2024) develop an Intelligent Random Forest model for logistics supply chain network risk prediction under uncertainty. The study integrates multi-source structured data and demonstrates strong predictive performance in classification tasks involving heterogeneous risk factors. The authors show that ensemble tree-based methods are well suited for modeling nonlinear relationships and complex feature interactions within supply chain systems.